

Official User Manual

GesturCTRL: Official User Manual

1. Introduction

GesturCTRL is a next-generation spatial computing utility that transforms your standard webcam into a high-fidelity hand-tracking sensor. It allows for seamless, touch-free control of your PC's cursor, volume, and system navigation using AI-driven landmark detection.

2. Installation & Setup

Prerequisites

- **Operating System:** Windows 10 or 11.
- **Hardware:** A standard 720p or 1080p webcam.

First-Time Launch

1. Run "**GesturCTRL.exe**"
 2. *Microsoft may stop the execution due to flagging it as an unrecognized app, however, click on "More Info" and then "Run Anyway" to run the app.*
 3. A "**Control Panel**" window will appear. Keep this window visible or minimized to maintain tracking.
 4. To close the application, press '**q**' on your keyboard while the Control Panel is the active window.
-

3. Gesture Dictionary

Action	Hand Gesture
Mouse Movement	Point your index finger at the screen. The cursor follows your fingertip.
Left Click	Perform a quick pinch (tap your thumb and index finger together).
Volume Control	Raise your pinky finger . While raised, the width of your pinch (thumb to index) adjusts volume from 0% to 100%.

Vertical Scroll	Raise your middle finger above the index finger level. Move it up to scroll up, and down to scroll down.
Task View	Raise your index, middle, and ring fingers simultaneously (Win+Tab shortcut).

4. Performance Optimization

- **Lighting:** Ensure your hand is well-lit from the front. Avoid standing directly in front of bright windows (backlighting).
- **Distance:** Position your hand **1.5 to 3 feet** away from the camera for optimal landmark recognition.
- **Background:** High-contrast backgrounds (solid walls) improve tracking stability compared to cluttered environments.

Privacy Policy

GesturCTRL: Privacy Policy

Effective Date: February 20, 2026

At **GesturCTRL**, I am committed to protecting your privacy. This policy explains how my application handles data, specifically your webcam feed and hand landmark information.

1. Local-Only Processing (The "Edge" Principle)

GesturCTRL is designed with a **privacy-first architecture**.

- **Zero Data Transmission:** All video processing, hand-tracking, and gesture recognition occur locally on your machine.
- **No Cloud Storage:** I do not upload, stream, or store your video feed or hand images to any external servers, databases, or third-party services.

2. Data Collection & Use

- **Webcam Access:** GesturCTRL requires access to your webcam solely to identify hand landmarks for gesture control.
- **Biometric Landmarks:** The application identifies 21 specific points on your hand (landmarks).

This data exists only in your computer's temporary memory (RAM) and is deleted instantly after each frame is processed.

- **Personal Information:** I do not collect names, email addresses, IP addresses, or any other personally identifiable information (PII).

3. Third-Party Libraries

GesturCTRL utilizes the following open-source frameworks:

- **MediaPipe (by Google):** Used for hand landmark detection.
- **OpenCV:** Used for video frame processing.
- **PyAutoGUI/Pycaw:** Used for system-level interaction (cursor and volume). *These libraries operate entirely within the local environment of the application.*

4. Security

Because the application does not connect to the internet to function, your data is as secure as your local computer. I recommend keeping your operating system and antivirus software up to date to protect your local environment.

License

MIT License

Copyright (c) [2026] [Ishaank Bansal]

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Disclaimer

Disclaimer:

GesturCTRL is a tool provided for convenience and productivity. By using this software, you acknowledge that I (the developer) am not responsible for any accidental clicks, data loss, or system changes caused by the gesture recognition system. Always ensure the "Control Panel" window is accessible so you can close the app if tracking becomes unstable.

CHANGELOG

GesturCTRL Version 1.1.0 (Under Development) [ETA: August 26]

- Integration of Lite Version [Backend: READY... Integration: PENDING]
- Refreshed UI (On/off switch, Version Toggle) [Backend: READY ... Integration: PENDING]
- Add sensitivity option [Backend: READY ... Integration: PENDING]

GesturCTRL Version 1.0.0 (Initial Release)

- Implemented 21-point hand landmark tracking via MediaPipe.
- Added smooth-factor cursor logic for stable navigation.
- Integrated master volume control and vertical scrolling.
- Added Window Management (Win+Tab) gesture.
- Bundled as a standalone Windows executable.